

Q1: $\|X\|_p = \left[\sum_i |x_i|^p \right]^{1/p}$

Prove it is a norm $\iff$ satisfies $\begin{cases} \textcircled{1} \text{ Positive} \\ \textcircled{2} \text{ Homogeneity} \\ \textcircled{3} \text{ Triangle Inequality} \end{cases}$

① Positive

$$\because |x_i|^p \geq 0$$

$$\therefore \|X\|_p = \left[\sum_i |x_i|^p \right]^{1/p} \geq 0 \quad (\|X\|_p = 0 \text{ if and only if } X = 0, \text{ each } x_i = 0)$$

② Homogeneity.

for any scalar $a \in \mathbb{R}$,

$$\|aX\|_p = \left[\sum_i |ax_i|^p \right]^{1/p} = \left[|a|^p \sum_i |x_i|^p \right]^{1/p} = |a| \left[\sum_i |x_i|^p \right]^{1/p} = |a| \|X\|_p$$

③ Triangle Inequality:

$$\text{To prove } \|x+y\|_p \leq \|x\|_p + \|y\|_p \quad (1)$$

Case A: if x or $y = 0$ obviously (1) is true.

Case B: when $\|x\|_p > 0$ and $\|y\|_p > 0$

~~Assume $\|x\|_p = \|y\|_p = 1$~~

$$\text{let } x_1 = \frac{x}{\|x\|_p}, \quad y_1 = \frac{y}{\|y\|_p}, \quad \lambda = \frac{\|x\|_p}{\|x\|_p + \|y\|_p}$$

$f(x) = \|x\|_p^p$ is convex

$$\therefore f(\lambda x_1 + (1-\lambda)y_1) \leq \lambda f(x_1) + (1-\lambda)f(y_1)$$

$$\Rightarrow \|\lambda x_1 + (1-\lambda)y_1\|_p^p \leq 1 \quad (f(x_1) < 1, f(y_1) < 1)$$

$$\Rightarrow \left\| \frac{\|x\|_p \cdot x}{\|x\|_p + \|y\|_p} + \frac{\|y\|_p \cdot y}{\|x\|_p + \|y\|_p} \right\|_p \leq 1$$

$$\Leftrightarrow \|x+y\|_p \leq \|x\|_p + \|y\|_p$$



Prove $DKL [p(x), q(x)] = \sum_{x \in X} p(x) \log \frac{p(x)}{q(x)} \geq 0$

Q2

$$X \in \{x_1, \dots, x_k\}$$

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① $p(x) = (p_1, \dots, p_k)$, $q(x) = (q_1, \dots, q_k)$

$$\sum p_i = 1, \sum q_i = 1$$

$$DKL [p(x), q(x)] = \sum_{i=1}^k p_i \ln \frac{p_i}{q_i}$$

② let $f(x) = \ln x$, $f(x)$ is concave, when $x > 0$

$$\text{let } x_i = \frac{q_i}{p_i} \Rightarrow \ln \frac{p_i}{q_i} = -\ln x_i = -f(x_i)$$

③ $f(x)$ is concave $\Leftrightarrow f(\sum p_i x_i) \geq \sum p_i f(x_i)$

$$\because \sum p_i x_i = \sum q_i = 1$$

$$\therefore \sum p_i f(x_i) \leq f(1) = 0$$

$$\sum p_i [-f(x_i)] \geq 0$$

(multiply -1 both sides)

according to ②'s result

$$\sum p_i \ln \frac{p_i}{q_i} \geq 0$$

$$\therefore DKL [p(x), q(x)] \geq 0$$

